System Components: Public Login, Student Dashboard, Admin Panel, Lessons, Assignment Upload, Assignment Review, Marks, Certificate Generation, Password Reset, Database, File Storage, API.

Tasks:

1. Identify at least 5 Assets

- Student personal information and profiles.
- Login credentials and tokens.
- Student marks and assessment feedback.
- Lessons and Learning materials.
- Database records.

2. Identify at least 4 Actors

- Student
- Lecturer
- Administrator
- Unauthenticated visitor

3. Identify at least 5 Entry Points

- Public Login form
- Password Reset
- Student Dashboard
- Admin Panel
- API endpoints.

4. Identify at least 3 Trust Boundaries

- Internet to Application
- Student to Administrator
- Application to File storage

5. Identify at least 5 Threats / Abuse Cases

- Uploading malicious file or malware
- Uploading file with disguised extensions
- Accessing another student's uploaded assignment.
- Uploading overloaded file to exhaust storage.
- Modifying certificates.

6. Recommend security controls for every threat

- MFA aka Multi-factor authentication.
- File validation, Size limits and so on.
- Role based access control.
- MFA and Least Privilege

7. Create a threat-model table

Asset	Actor	Entry Point	Threat	Security Control
Student accounts	Attacker	Login	Password guessing and account takeover	MFA, strong password hashing and rate limiting
Assignment files	Malicious student	Assignment Upload	Uploading malicious files	File validation, size limits and malware scanning
Student marks	Student	Dashboard / API	Accessing or modifying another student's marks	Role-based access control and authorization checks
Admin panel	Attacker	Admin Panel	Unauthorized administrative access	MFA and least privilege
Certificates	Malicious student	Certificate Generation	Forging or modifying certificates	Server-side generation and certificate verification
Password reset	Attacker	OTP endpoint	OTP guessing and account takeover	Short-lived OTPs and rate limiting
Database	Attacker	API	SQL injection and unauthorized data access	Parameterized queries and access controls

8. Review the Assignment Upload feature

Potential threats:

- Uploading malicious files or malware.
- Uploading oversized files to exhaust storage.
- Uploading files with disguised extensions.
- Accessing another student's uploaded assignment.

Recommended controls:

- Allow only approved file types.
- Validate file extensions and actual file signatures.
- Apply file-size limits.
- Scan uploaded files for malware.
- Store files securely and enforce authorization before downloads.

9. Review the Certificate Generation workflow

Potential threats:

- Students generating certificates without completing requirements.
- Modifying certificate details or marks.
- Downloading another student's certificate.
- Generating duplicate or fraudulent certificates.

Recommended controls:

- Verify student eligibility using trusted database records.
- Generate certificates on the server.
- Restrict certificate generation to authorized users.
- Assign unique certificate IDs.
- Provide a certificate verification mechanism.

10. Review Password Reset / OTP abuse risks

Potential threats:

- OTP brute-force attacks.
- Reusing expired OTPs.
- Requesting too many OTPs.
- Discovering registered accounts through error messages.
- Unauthorized password changes.

Recommended controls:

- Use random, short-lived, single-use OTPs.
- Limit OTP verification attempts.
- Apply rate limiting.
- Use generic error messages.
- Notify users after successful password resets.